

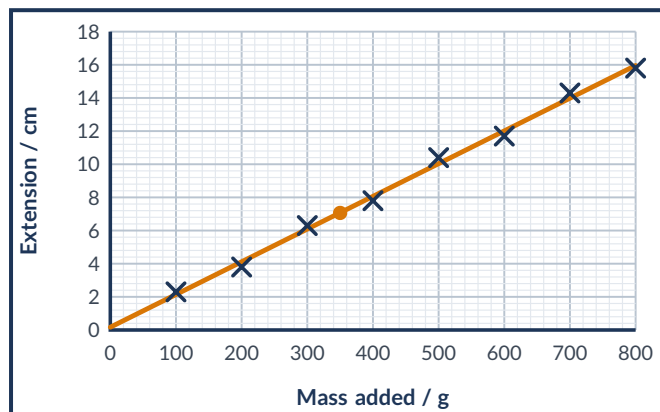
TEACHER COPY

Lesson 5 worksheet: answers

Orange lines are model answers; accept any sensible line that balances the points.

1 A straight line

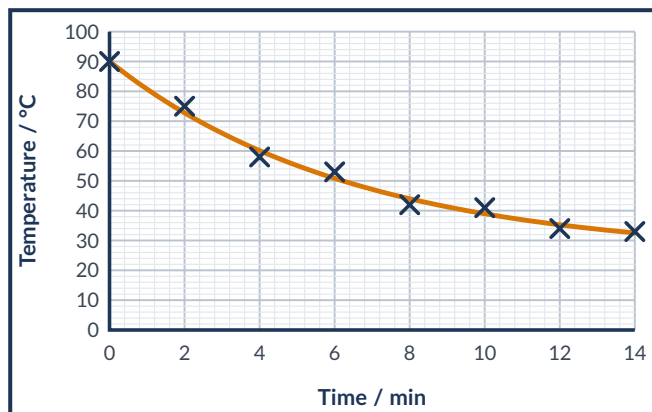
SPRING



a About 7 cm (accept 6.5–7.5), read from the line, not a point.
b About 4 / 4; the exact split matters less than the balance.
 Ruler, one thin line, no zig-zag.

2 A curve

COOLING



a About 55 °C (accept 52–58). **b** The points fall quickly then level off, so a straight line would sit above the points in the middle and below them at the ends. Cooling slows as the water gets nearer room temperature.

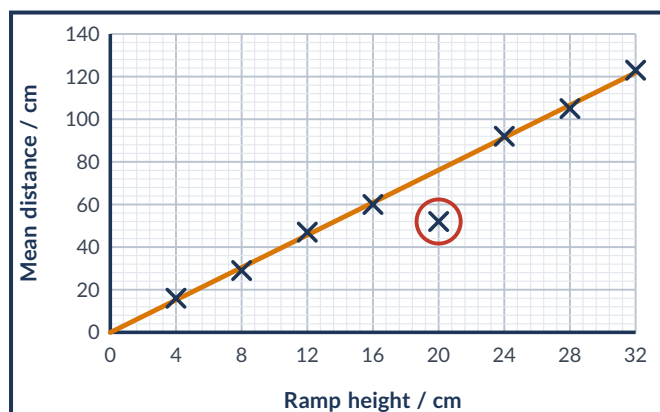
3 Good line or not?

JUDGE IT

C only is good: a straight line with the points balanced either side. **A** is dot-to-dot, so it treats every measurement as exact; **B** is too shallow, so the points drift above the line and miss the trend; **D** is a wobbly curve through every point, which follows the scatter instead of the trend.

4 The point that does not fit

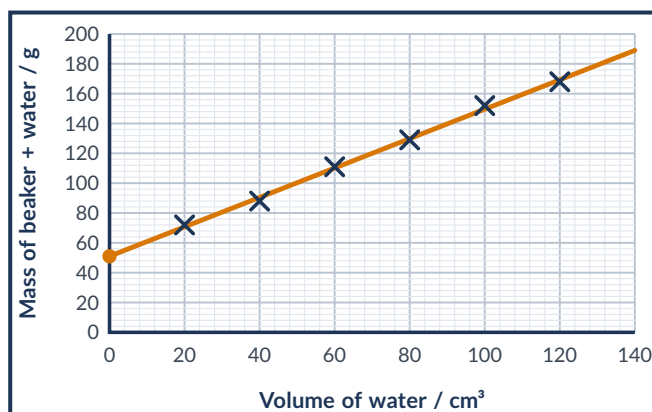
OUTLIER



a Outlier at (20, 52); the line follows the other seven points, so it passes well above it. **b** Neither: do not delete it or bend the line to it. First check the record, then repeat the 20 cm run. **c** A repeat near 76 cm suggests a measuring or release mistake: note it on the graph and use the repeat. **d** If it stays near 52 cm, it may be real physics (a bump or a sticking wheel). Report it, investigate, and do not hide it.

5 Directly proportional?

THINK



a About 50 g (accept 45–55). **b** About 90 g and 130 g: 130 is not double 90, so no. **c** No. The line is straight but does not pass through the origin. **d** The intercept is the mass of the empty beaker. Zero the balance with the beaker on (tare it), or subtract 50 g from every reading; then the line passes through (0, 0).

Talk about it: an outlier is left out of the *line* only when there is a reason, such as a recorded mistake, and it stays circled on the graph. A straight line with an intercept still shows a linear pattern and supports predictions; it just is not proportional, and the intercept often points to a systematic error or an extra fixed quantity.